



An improved kinetic mechanism for 3-pentanone pyrolysis and oxidation developed using multispecies time histories in shock-tubes



Enoch E. Dames*, King-Yiu Lam, David F. Davidson, Ronald K. Hanson

High Temperature Gasdynamics Laboratory, Department of Mechanical Engineering, Stanford University, United States

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ABSTRACT

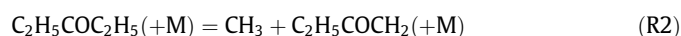
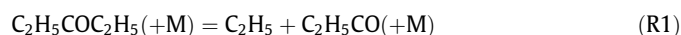
Laser-based OH, CO, CH₃, and H₂O species histories during 3-pentanone oxidation were measured behind reflected shock waves over the temperature range of 1277–1678 K at pressures around 1.6 atm, and for equivalence ratios of 0.5, 1.0 and 1.5, complementing recent work also conducted in this laboratory (Lam et al., 2012). These data are used to develop an improved detailed kinetic mechanism for 3-pentanone pyrolysis and oxidation. A 3-pentanone submechanism consisting of 13 species and 61 reactions is superimposed upon the H₂/CO/C₁–C₄ base mechanism of JetSurF 2.0 (Wang et al., 2010), with several reactions updated in order to obtain better agreement with shock tube oxidation data. The improved mechanism shows satisfactory agreement with the wealth of shock tube data presented here and previously published, although there remain some conditions where the model can benefit from further improvements. Major rate parameter uncertainties in the model are also discussed and will be used in a companion work addressing model optimization and experimental design of shock tube studies using data obtained with the Stanford shock tube.

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1. Introduction

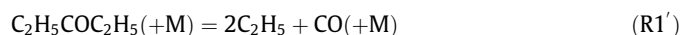
An accurate understanding of the combustion kinetics for 3-pentanone is important for two major reasons. The first is that this molecule is used as a fuel tracer in non-invasive laser-induced fluorescence measurements of both temperature and exhaust gas recirculation under combustion conditions and within internal combustion engines [1–4]. The second is that 3-pentanone represents an archetypical small oxygenated fuel with intermediate radical species (e.g., C₂H₅CO/propionyl, CH₃CHCO/methyl ketene) that are expected to exist during the combustion of other bio-derived alternative fuels, including secondary alcohols.

3-Pentanone also lends itself well to kinetic modeling because its symmetry considerably reduces the number of reactions necessary to describe its pyrolysis and oxidation in comparison with its isomers. Two pathways dominate its thermal decomposition (also depicted in Fig. 1):



The C–C bonds nearest the carbonyl group have been estimated to be 82.3 ± 1.2 kcal/mol [5] while the terminal methyl groups are expected bound with similar energy, to within 1 kcal/mol, based

on analogous bond energy differences in propanal [5]. Thus, differences in its branching ratio for decomposition are expected to be primarily entropic in origin. Under the pyrolytic conditions of this study, the propionyl radical resulting from R1 (C₂H₅CO) very quickly decomposes into the C₂H₅ radical and CO, while the C₂H₅COCH₂ radical in R2 will decompose to ketene and the C₂H₅ radical. At high enough temperatures, R1 can be represented by



Shock tube measurements of CO mole fraction therefore provide an excellent constraint on the rate of this reaction. However, ethyl decomposition to ethylene + H is in competition with the chemically activated C₂H₅ + H recombination followed by dissociation to two methyl radicals. Even though ethyl decomposition is fast under the conditions of this study (≤ 10 μs), it is still orders of magnitude slower than propionyl decomposition to CO + C₂H₅ [6].

A detailed kinetic mechanism for the oxidation and pyrolysis of 3-pentanone was adapted from the work of Serinyel et al. (referred to here as the [NUI Galway] C³ mechanism) [7]. This mechanism was used as a starting point by Lam and coworkers [8] as well as recent work by Dames et al. [9]. Although a different base mechanism was chosen here and in our recent work [9], many of the same reactions belonging to the 3-pentanone submechanism were retained, with details to follow. One motivation for the present work was the realization of a decomposition pathway in the C³

* Corresponding author.

E-mail address: edames@stanford.edu (E.E. Dames).

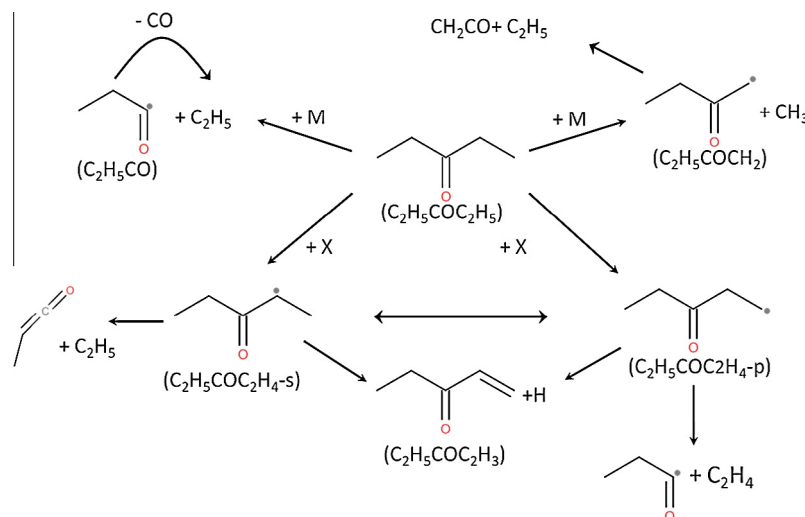
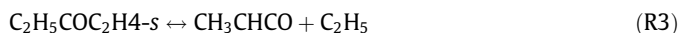


Fig. 1. Reaction pathway diagram illustrating the initial dominant consumption pathways during the combustion of 3-pentanone. Not shown are further reactions of $C_2H_5COC_2H_3$, $X = H, OH, HO_2, CH_3$, etc.).

mechanism with erroneously high rates [7]. That is, the unimolecular decomposition reaction



where CH_3CHCO is methyl ketene, was too fast by several orders of magnitude under the temperature range of this work. This resulted in a high flux of 3-pentanone to methyl ketene, with subsequent methyl ketene loss primarily occurring through bimolecular reaction with other gas phase radical species. In R3, *s* refers to a secondary alkyl radical site (see Fig. 1; *p* refers to a primary alkyl radical site).

In the context of 3-pentanone as a fuel tracer, its rate of decomposition as a function of temperature is important in interpreting its fluorescence signal strength and extracting information regarding temperature and exhaust gas recirculation mole fractions within internal combustion engines. The central purpose of this work is to present a 3-pentanone mechanism that can be reliably used to determine its lifetime under engine-relevant conditions. Additionally, the 3-pentanone shock tube oxidation data proved useful in highlighting the importance of smaller hydrocarbon chemistry in accurately describing the decomposition of 3-pentanone to combustion products. Updates to the core chemistry of JetSurF resulted in significantly improved agreement between this updated model and 3-pentanone oxidation data. The 3-pentanone submechanism was adapted from recent work [7,8], with minor changes detailed below. This work will also be accompanied by a companion paper in which the improved 3-pentanone mechanism presented here, along with our wealth of shock tube data, will be used to test an experimental design methodology [10] to determine the minimal basis set of experimental data needed to constrain the model rate parameter uncertainties to the same degree as the full experimental data set.

2. Methodology

2.1. Shock tube studies

In the present study, OH, CO, CH_3 , and H_2O species histories during 3-pentanone oxidation were measured behind reflected shock waves over the temperature range of 1277–1678 K at pressures around 1.6 atm, and for equivalence ratios of 0.5, 1.0 and 1.5, complementing recent work also conducted in this laboratory

[8]. Details regarding the shock tube setup and laser absorption diagnostics can also be found elsewhere [8]. The gas mixture compositions and measured species for the previous and current studies are listed in Table 1. Chemkin II [11] was used to model the shock tube conditions (under constant internal energy and volume constraints) found in Table 1.

As previously discussed [8], 2σ uncertainties in the methyl concentration profiles are approximately $\pm 10\%$ in the first 100 μs , which is mainly contributed from the uncertainty in the absorption coefficient and the interference subtraction scheme. At later times, the CH_3 plateau level has a slightly larger 2σ uncertainty of $\pm 20\%$ due to larger interference absorption (primarily from C_2H_4). Measured mole fractions of other species (3-pentanone, CO, C_2H_4 , OH, and H_2O) are primarily dependent on the absorption coefficient which is temperature sensitive. 2σ uncertainties in 3-pentanone, CO, C_2H_4 , and H_2O species profiles are $\pm 10\%$, and $\pm 5\%$ for OH (e.g., see Ref. [38] and Table 2 therein). The 2σ uncertainty in the measured post-shock temperature is $\pm 1.0\%$ and is mainly due to the uncertainty in the measured shock velocity ($\pm 0.2\%$) [8]. The uncertainty in temperature becomes especially relevant when comparing model predictions with some measured species profiles (e.g., those that can be used to define ignition delay times), and is further discussed below.

Thermodynamic parameters for all species in the 3-pentanone submechanism were computed using ther gas, which uses group-additivity based estimates to compute thermodynamic properties of chemical compounds [12]. Thermodynamic properties for some radical intermediates were found to be in considerable disagreement with those estimated in the work of Serinyel et al. [7] (see Table 2). Thus, the standard 298 K formation enthalpies of several radical species ($C_2H_5COC_2H_4-s$, $C_2H_5COC_2H_4-p$, and C_2H_5CO) were also computed by means of isodesmic reactions at the M06-2X/6-311+G(d,p) level of theory [13] and found to be within 10% of ther gas estimates made in the present work. The isodesmic work reactions and details can be found in the Supplemental information. Because the experimental formation enthalpies of some species needed for isodesmic reactions of all 3-pentanone radical intermediates are not available, all thermodynamic parameters of 3-pentanone radical intermediates are utilized from ther gas in order to retain consistency among these properties in the 3-pentanone submechanism. Nevertheless, ther gas estimates are in good agreement with the isodesmic-derived formation enthalpies. Corresponding NASA polynomials are also provided in the

Table 1

Summary of test gas mixture compositions and measured species used as targets.

Mixtures	Gas compositions (in Argon) ^a		Species time-histories					
	3-Pentanone (%)	O ₂ (%)	3-Pentanone	CH ₃	CO	C ₂ H ₄	OH	H ₂ O
A	1.000	–	×			×		
B	0.250	–			×			
C	0.100	–		×				
D	0.040	0.280					×	×
E	0.040	0.560					×	×
F	0.040	0.187						×
G	0.040	0.560			×			×
H	0.075	0.525		×			×	×

^a Mixtures A–E from [8]; F–H: this work.**Table 2**Comparisons of $\Delta_f H_{298K}^0$ (kcal/mol) for selected species in the 3-pentanone submechanism. See supporting information for NASA polynomial formatted thermodynamic data.

Species ^a	This work	Serinyel et al.	Isodesmic ^b	Experimental
3-Pentanone	–61.9	–62.2		–61.5 ^c
C ₂ H ₅ COC ₂ H ₄ -s	–21.7	–18.3	–23.0	
C ₂ H ₅ COC ₂ H ₄ -p	–13.1	–13.2	–11.7	
C ₂ H ₅ COC ₂ H ₃	–29.1	–35.7	–27.0	
C ₂ H ₄ COC ₂ H ₃ -p	19.7	13.3		
C ₂ H ₄ COC ₂ H ₃ -s	11.1	4.1		
C ₂ H ₄ COC ₂ H ₂	29.7	23.4		
C ₂ H ₅ CO	–9.8	–8.0	–9.2	
C ₂ H ₃ CO	18.6	11.6		

^a See Fig. 1 for corresponding Kukulé structures.^b See supporting information for details.^c Average of three values: –60.6 ± 0.20 [15], –62.25 ± 0.39 [16], –61.65 ± 0.20 [17] (note: the average does not necessarily represent a more accurate representations of the quoted value).

Supplemental information. Lastly, thermodynamic properties were updated with values in the most recent Burcat compilation [14] for the following species: CO, CO₂, HCO, H₂CO, CH₃, C₂H₄, and C₂H₅. These updates resulted in a noticeable improvement between model predictions and the shock tube oxidation data.

In an effort to identify major sources of simulated-experimental data discrepancies, 2σ uncertainties were assigned to the pre-exponential factors of all reactions in the 3-pentanone submechanism, while those for the base mechanism are taken from Sheen et al. [18]. In general, pressure-dependent reactions were assigned uncertainty factors ranging from 3 to 5 due to greater uncertainty in the falloff region for many of the reactions here. A full list of the 3-pentanone reactions and their attendant uncertainty parameters may be found in the Supplemental information.

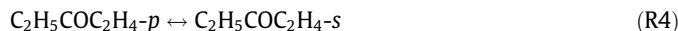
2.2. Rate parameter changes and updates

As mentioned in the introduction, the initial detailed kinetic mechanism for the oxidation and pyrolysis of 3-pentanone was adapted from the C³ mechanism [7]. This mechanism was used as a starting point by Lam and coworkers [8]. Although a different base mechanism was chosen here, some of the same reactions belonging to the 3-pentanone submechanism were retained. However, many of the unimolecular decomposition reactions were necessarily redefined, with details below.

R1 and R2 were initially adopted from the work of Serinyel et al. [7], who estimated the reverse reactions based on recommendations for ethyl recombination with acetyl and methyl recombination with *n*-propyl, respectively. Rates in the falloff regime were then determined through QRRK with a master equation analysis [7]. These rates were adopted with the exception of two changes.

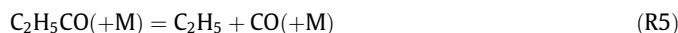
The reverse of R2 was more appropriately assigned as the recombination rate of methyl with allyl (*a*C₃H₅) to 1-butene [19], due to the similarities in resonance between allyl and the C₂H₅COCH₂ radical. Secondly, R1 and R2 were increased by a factor of 3.2, similar to our previous work [8,9]. This is further discussed below.

The two H-abstraction rates from 3-pentanone by H were decreased by 25% in order to improve agreement between prediction of early OH plateaus for oxidation and agreement between 3-pentanone pyrolytic decay at later times. All other bimolecular H-abstraction reactions from C₂H₅COC₂H₅ were adopted from the C³ mechanism and assigned uncertainty factors of 2. The H-shift reaction



was not included in previous work [7] although its inclusion is expected to be important since the isomerization reaction is fast under the conditions of this work and equilibrium favors the more stable (by 9 kcal/mol at 298 K) secondary radical. The rate of R4 was estimated based on recent theoretical work by Sirjean and coworkers [20] with an estimated uncertainty factor of 3. The unimolecular decomposition of C₂H₅COC₂H₄-s has two possibilities, depicted in Fig. 1. Decomposition of C₂H₅COC₂H₄-s to C₂H₅+methyl ketene was estimated in the reverse direction by analogy to the recombination of ethyl with ketene [21]. The resulting methyl ketene unimolecular decomposition was estimated in the reverse reaction by analogy with triplet methylene + CO recombination to ketene, with an estimated uncertainty factor of 5. Hydrogen elimination from C₂H₅COC₂H₄-s was estimated by analogy H-elimination in 1-butene-3-yl to 1,3-butadiene + H [22,23], and results in the further unsaturated stable intermediate C₂H₅COC₂H₃. Hydrogen elimination from C₂H₅COC₂H₄-p was not included in the work of Serinyel et al. and therefore included in this work, estimated by analogy to 1-butene-4-yl = 1,3-butadiene + H [23]. All bimolecular H-abstraction reactions from C₂H₅COC₂H₃ by H, OH, O, etc., were adopted from the C³ mechanism and assigned uncertainty factors of 2. Reactions involving radical derivatives of C₂H₅COC₂H₃ were estimated from analogous reactions in the C₂H₅COC₂H₅ class of reactions. Under all of the conditions of this study, the predicted concentrations of C₂H₅COC₂H₃ did not exceed several ppm, thus the measured species profiles are insensitive to the kinetics of C₂H₅COC₂H₃.

The rate parameters for the decomposition of C₂H₅CO (propionyl radical)



were partially adopted from the work of Kerr and Lloyd [24]. Because the aforementioned authors determined only *k_o* and *k_∞*, rates in the falloff region were estimated by assuming the same collisional energy transfer characteristics as in acetyl/CH₂CHO decomposition [23]. Mereau and coworkers [25] used a combination of semi-empirical quantum chemical methods, in addition to transi-

tion state theory, to estimate the high-pressure limit rate constant for this reaction, but the rates are between a factor of 5 and 6 higher than those recommended by Kerr and Lloyd for the temperatures of this work. In addition, the corresponding rates in the C^3 mechanism are a factor of 2–4 slower than that of Kerr and Lloyd, although the source of this rate is unclear and the rate parameters suffer from lack of pressure dependence. Preliminary theoretical calculations for R5 were also performed as part of the allyloxy decomposition system [26]. More advanced calculations for this particular reaction were performed by Goldsmith [27]. Geometry optimization and normal mode analysis were done at the CCSD(T)/cc-pvtz level theory. Next, a sequence of corrections were applied to obtain more accurate energies. First, the CCSD(T) energy was extrapolated to the complete basis-set limit using augmented quadruple and quintuple basis sets. Next, corrections for higher-order excitations were computed from the difference between CCSDT(Q)/cc-pvdz and CCSD(T)/cc-pvdz. Core-valence correlations were determined from the difference in CCSD(T) energy if the core electrons were frozen or not, extrapolated from quadruple and triple basis sets. Lastly, relativistic effects were determined from the difference in configuration interaction energy with and without the Douglas-Kroll one-electron integrals. The $2\text{-}\sigma$ uncertainty in the electronic energy at 0 K is less than 0.3 kcal/mol. All calculations were done in MOLPRO [28] using the Dunning-style correlation-consistent basis sets. The temperature- and pressure-dependent rate constants were computed from RRKM/Master Equation theory, as implemented in VariFlex [29]. A single-exponential down model for collisional-energy transfer was used, with $\langle \Delta E \rangle_{\text{down}} = 250 (T/298)^{0.85} \text{ cm}^{-1}$. The uncertainties in the underlying electronic structure properties and the parameters for collisional energy transfer were propagated into the final rate coefficients. The resulting uncertainty in the rate coefficients was a factor of two. The resulting rate calculations at 1.6 atm and between 800 and 1600 K are only 10–22% higher than the current estimate. Comparisons of selected propionyl decomposition rates at 1.6 atm are illustrated in Fig. 2. Under the conditions of this work, propionyl undergoes nearly instantaneous decomposition to $\text{CO} + \text{C}_2\text{H}_5$. Propionyl decomposition channels leading to methyl ketene/ $\text{CH}_3\text{CHCO} + \text{H}$ and $\text{CH}_2\text{CO} + \text{CH}_3$ were also explored and estimated by analogy to $\text{CH}_2\text{COH} (+\text{M})$:

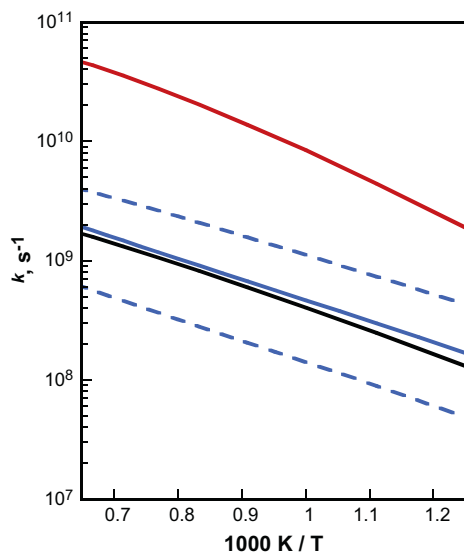


Fig. 2. Comparison of thermal propionyl ($\text{C}_2\text{H}_5\text{CO}$) decomposition rates at 1.6 atm and between 800 and 1600 K. Solid black line: this work (see text for details); solid blue line: RRKM/ME results of Goldsmith with estimated uncertainty band [27]; solid red line: Serinyel et al. (k_∞ only) [7]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

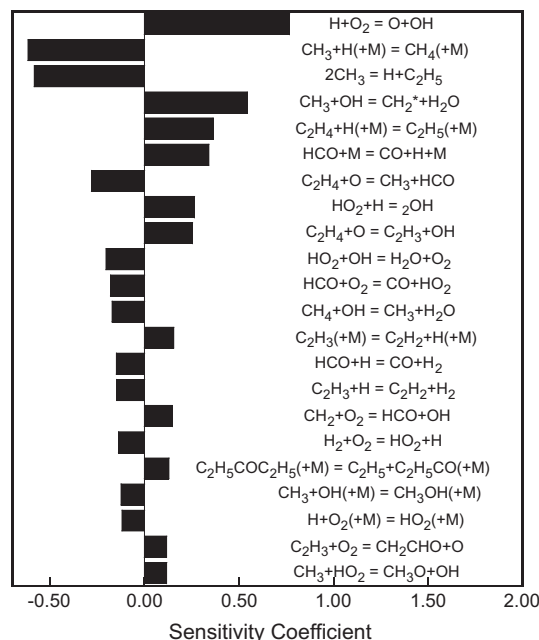


Fig. 3. Sensitivity of OH concentration to reaction rate parameters in JetSurF 2.0 [23] at 1217 K, 1.74 atm, and at 1200 μs after arrival of the shock wave.

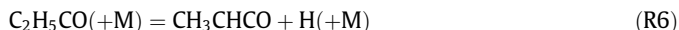
Table 3

List of reactions added or changed from the core chemistry of JetSurF 2.0 [23].

Reaction	JetSurF 2.0	This work
$\text{H} + \text{O}_2 = \text{O} + \text{OH}$	[41]	[42]
$\text{HO}_2 + \text{OH} = \text{H}_2\text{O} + \text{O}_2$	[43]	[44]
$\text{HCO} + \text{OH} = \text{H}_2\text{O} + \text{CO}$	[45]	[36]
$\text{HCO} + \text{O}_2 = \text{HO}_2 + \text{CO}$	[46]	[47]
$\text{HCO} + \text{M} = \text{CO} + \text{H} + \text{M}$	[48] ^a	[48]
$\text{CH}_2 + \text{O}_2 = \text{OH} + \text{H} + \text{CO}$	[41]	[40]
$\text{CH}_2 + \text{O}_2 = 2\text{H} + \text{CO}_2$		
$\text{CH}_2 + \text{O}_2 = \text{O} + \text{CH}_2\text{O}$		
$\text{CH}_2 + \text{O}_2 = \text{H}_2 + \text{CO}_2$		
$\text{CH}_2 + \text{O}_2 = \text{H}_2\text{O} + \text{CO}$		
$\text{CH}_3 + \text{OH} (+\text{M}) = \text{CH}_3\text{OH} (+\text{M})$	[41]	[35]
$\text{CH}_3 + \text{OH} = \text{CH}_2 + \text{H}_2\text{O}$	[41]	[35]
$\text{CH}_3 + \text{OH} = \text{CH}_2^* + \text{H}_2\text{O}$	[41]	[36]
$\text{C}_2\text{H}_4 + \text{O} = \text{CH}_3 + \text{HCO}$	[49]	[36]
$\text{C}_2\text{H}_4 + \text{O} = \text{H} + \text{CH}_2\text{CHO}$	[49]	[37]
$\text{C}_2\text{H}_4 + \text{O} = \text{CH}_2 + \text{CH}_2\text{O}$	[49]	[37]
$\text{C}_2\text{H}_4 + \text{OH} = \text{C}_2\text{H}_3 + \text{H}_2\text{O}$	[50]	[38]
$\text{C}_2\text{H}_4 + \text{OH} = \text{CH}_2\text{O} + \text{CH}_3$		[39]
$\text{C}_2\text{H}_4 + \text{OH} = \text{H} + \text{CH}_3\text{CHO}$		
$\text{C}_2\text{H}_4 + \text{OH} = \text{H} + \text{C}_2\text{H}_3\text{OH}^b$		
$\text{C}_2\text{H}_2 + \text{H} (+\text{M}) = \text{C}_2\text{H}_3 (+\text{M})$	[51]	[36]

^a Increased by a factor of 2 in JetSurF and restored to nominal value in this work.

^b Assigned as acetaldehyde in the detailed mechanism.



However, these channels do not compete with the $\text{CO} + \text{C}_2\text{H}_5$ channel and were not included in the final present mechanism.

Regarding changes and updates to the base mechanism, the rate of ethyl decomposition to ethylene + H



was reduced by 40% from its nominal value (R8 has an estimated uncertainty of 50% [30,31]) This reduction was performed to better fit methyl time histories in 3-pentanone pyrolysis and initial OH rise under oxidative conditions, and is further discussed below.

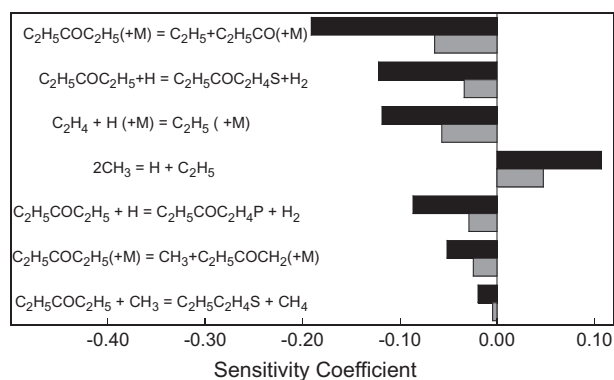


Fig. 4. Sensitivity of 3-pentanone concentration during pyrolysis to reaction rate parameters in the current model [23] at 1323 K, 1.6 atm, and at 25 μ s (gray bars) and 200 μ s (black bars) after arrival of the shock wave.

Such reductions have been implemented elsewhere [7,32–34] and are typically performed so that the rate of the reverse reaction is slower for $T > 1100$ K where experimental data are scant. We note that, during the course of this work, an error in the JetSurF [23] and USC Mech II [52] mechanisms were revealed. This error is in R8, and is due to the last two Troe parameters being of incorrect sign. This results in a rate constant too fast by over a factor of two for the conditions of this work. Specifically, the Troe parameters are incorrectly reported as $\alpha = -0.569$, $T^{**} = 299$, $T^* = 9147$, and $T^* = -152.40$ (see Dames and Golden [53] for a description of the formalisms of Troe [54,55]). The correct parameters as reported by Miller and Klippenstein [30] are in fact: $\alpha = -0.569$, $T^{**} = 299$, $T^* = -9147$, and $T^* = 152.4$.

Numerous updates were also made to the base JetSurF 2.0 [23] core $H_2/CO/C_1-C_4$ chemistry. The sensitivity spectra for [OH],

shown below in Fig. 3, illustrates what reactions dominate the nearly steady OH concentration shown in Fig. 8 between 100 and 1000 μ s for the 1217 K OH profile. Sensitivity analyses in these areas served to guide updates to the base model. Notable changes include the recent theoretical work by Jasper et al. [35] for the activated methanol system, while $CH_3 + OH = CH_2 + H_2O$ was adopted from the most recent Baulch compilation [36]. Because significant amount of ethylene is produced during 3-pentanone combustion, several reactions of ethylene with radical species were also updated [36–39]. Five reactions of triplet methylene with molecular oxygen were adopted from Bley and coworkers [40] and found to substantially improve model prediction agreement with the shock tube oxidation data. Table 3 below displays all reactions that were updated or added to JetSurF 2.0. As can be seen from Figs. 7 and 8, the above changes to the base mechanism resulted in significantly improved agreement with OH and H_2O profiles.

3. Results and discussion

3.1. 3-Pentanone pyrolysis

Fig. 4 shows a sensitivity spectra for 3-pentanone decay modeled at 1323 K, 1.6 atm, and at both 25 and 200 μ s after arrival of the reflected shock. 3-pentanone decay is sensitive to a large number of reactions in contrast to our previous work [8] which utilized a veritably different mechanism and where the rate of 3-pentanone decay was almost exclusively sensitive to its own unimolecular decomposition. Using the improved and corrected mechanism however, 3-pentanone decay is revealed to be sensitive to more than just its own decomposition. The unimolecular decomposition of 3-pentanone to ethyl and propionyl tops the list because propionyl quickly decomposes into $C_2H_5 + CO$. Thus, the net reaction of this decomposition channel is represented by R1' and is a large

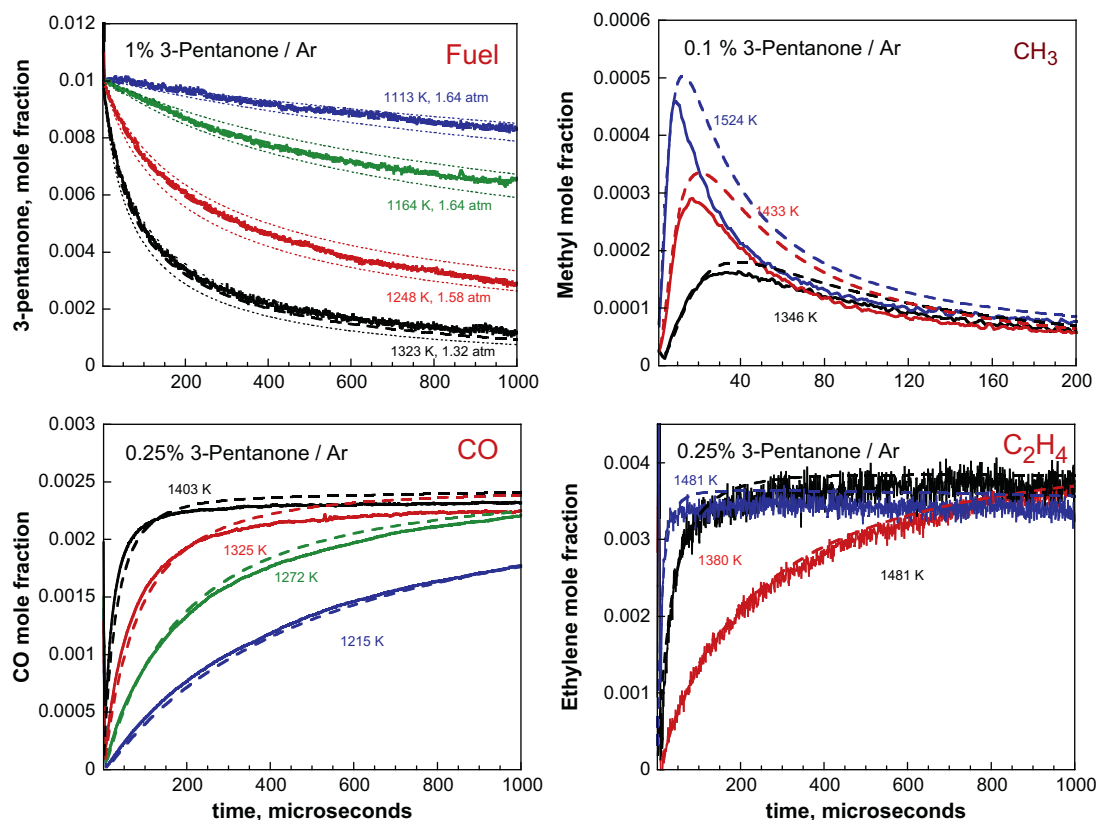


Fig. 5. Comparison of present 3-pentanone mechanism predictions (dashed lines) with selected experimental pyrolysis species profiles [8]. The dotted lines in the 3-pentanone profile plot illustrate simulation results with a $\pm 1\%$ change from the nominal temperature (equivalent to the measured temperature uncertainty).

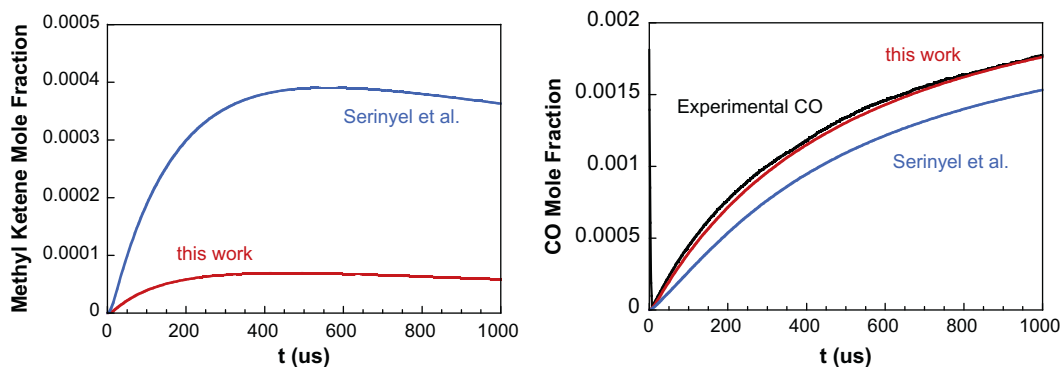


Fig. 6. Comparisons of methyl ketene (left) and CO (right) mole fractions for 0.25% 3-pentanone in Ar, $T_5 = 1215$ K, $P_5 = 1.68$ atm. Red lines: this work. Blue lines: this work with k_3 from the model of Serinyel et al. [7]. Black line: Measured CO mole fraction [8]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

source of reactive ethyl radicals. As elaborated upon in our earlier work [8], this explains why large amounts of both CO and C_2H_4 are produced during 3-pentanone pyrolysis and oxidation. The abundance of C_2H_5 immediately following 3-pentanone decay results in high sensitivities for not only its decomposition to ethylene + H, but also the chemically activated reaction of $C_2H_5 + H = 2CH_3$. In the current mechanism, this reaction is taken from GRI Mech 3.0 [41].

Fig. 5 displays model-data comparisons for 3-pentanone pyrolysis behind reflected shock waves at various conditions. Our previous work showed that the unimolecular decomposition rates of 3-pentanone in the work of Serinyel et al. [7] were too slow by over a factor of three [8]. Recall that in this work, two modifications were

made. The reverse of R2 was assigned as the recombination rate of methyl with allyl (aC_3H_5) to 1-butene [19], due to the similarities in resonance between allyl and the $C_2H_5COCH_2$ radical. The result of this change is that the 3-pentanone branching ratio, k_1/k_2 , was altered. At 1.6 atm, this ratio ranges from 77% and 54% between 1000 K and 1600 K respectively. In contrast, this corresponding range is 40–50% in the work of Serinyel et al. [7]. At lower temperatures, the net effect is quite large since R1 is such a fast source of H radicals. The subsequent fit of the total 3-pentanone decay rate to the 3-pentanone pyrolysis data resulted in an increase by a factor of 3.2. Between 1200 and 1600 K, these absolute rates are within our previously reported uncertainty of $\pm 35\%$ in the total decay rate of 3-pentanone [8]. As seen in Fig. 5, the 3-pentanone decay

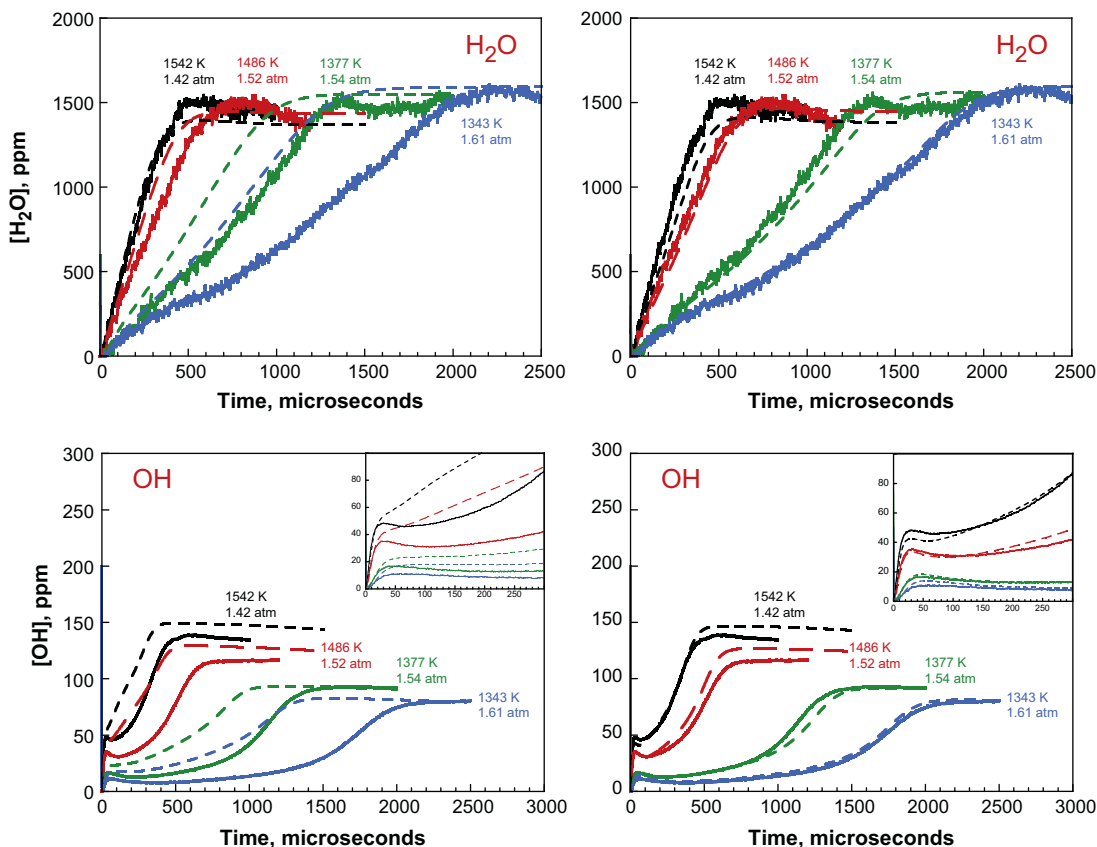


Fig. 7. Comparisons of measured and simulated H_2O and OH time-histories for 400 ppm 3-pentanone with 2800 ppm O_2 in Ar ($\phi = 1.0$). Left figures show comparisons with model prior to updating core chemistry (see text for details). Inset figure provided for OH to show the early-time features over 0–300 μs .

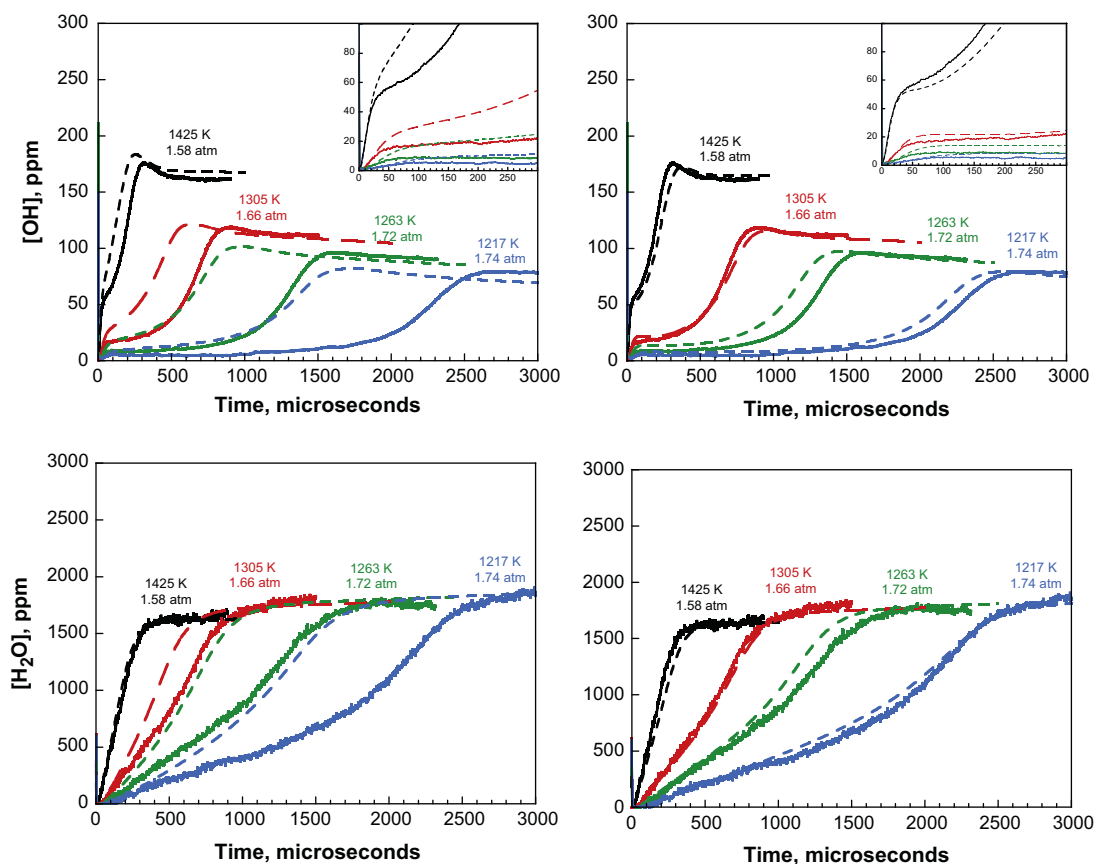


Fig. 8. Comparisons of measured and simulated H_2O and OH time-histories for 400 ppm 3-pentanone with 0.56% O_2 in Ar ($\phi = 0.5$). Left figures show comparisons with model prior to updating core chemistry (see text for details). Inset figures are provided for OH to show the early-time features over 0–300 μs .

is captured very well. Early methyl rise is also captured well, although the peaks are slightly overpredicted and the disappearance is not fully captured. As discussed in our previous work [9], this discrepancy can be resolved with small modifications to relatively uncertain methyl consumption pathways. This was not attempted here, as a full mechanism optimization will be addressed in the near future. Also shown in Fig. 5 are comparisons between simulated results and CO and C_2H_4 profiles. Overall, the agreement between the model and all pyrolysis data are excellent, although both CO and C_2H_4 plateaus are slightly overpredicted. The simulated early CO formation rate is also too slow, although still within the 10% uncertainty of the experimental data. The slight underprediction of early CO suggests that branching of 3-pentanone decomposition should more heavily favor R1. Thus, reliable quantum chemical calculations of the 3-pentanone potential energy surface in conjunction with reliable reaction rate theory calculations are needed to resolve this issue. These calculations will be addressed in future work. Regardless, model performance against the pyrolysis targets in Fig. 5 is excellent considering the large uncertainties in some of the reactions that contribute to the observed features in the species time histories.

As mentioned in the introduction, the decomposition of $\text{C}_2\text{H}_5\text{COC}_2\text{H}_4$ -s (R3) in the work of Serinyel et al. is too high. R3 was inadvertently given rate parameters that corresponded to the rate of the reverse reaction, resulting in a decomposition rate too high by several orders of magnitude. This resulted in a simulated excess of methyl ketene, as illustrated in Fig. 6. This error was not discovered in our own recent work [8], and as a result, a methyl ketene decomposition pathway was introduced. This pathway was also assigned an erroneously high decomposition rate due to the improper assumption that the rate parameters for the reaction

$\text{C}_2\text{H}_4 + \text{CO} \rightarrow \text{CH}_3\text{CHCO}$ could be estimated by those for the reaction of $\text{CH}_2 + \text{CO} \rightarrow \text{CH}_2\text{CO}$. This error led to the conclusion that methyl ketene unimolecular decomposition is necessary in 3-pentanone kinetics. Indeed, with a proper estimation of methyl ketene decomposition rate, simulation results illustrate that its decay through unimolecular decomposition is negligible and primarily occurs through bimolecular reaction with gas phase radical species. With these past revelations recognized and with corresponding corrections made, excellent agreement between the modeled and experimental CO profiles is achieved (see Fig. 6).

3.2. 3-Pentanone oxidation

Figs. 7 and 8 below show H_2O and OH profiles for varying shock conditions [8] and at equivalence ratios of 1.0 and 0.5, respectively. The improvements in model-data agreement obtained by updating the core chemistry are clear and significant. As previously noted [8], low fuel concentration mixtures are preferred in species time-history measurements during hydrocarbon oxidation in order to minimize the rapid energy release at the time of ignition, which increases the reflected shock temperature. This rapid heat release additionally changes the absorption coefficients of OH and H_2O . For 3-pentanone shock tube studies where an initial fuel concentration of 400 ppm is used, an approximate ~ 50 – 60 K increase in temperature is simulated. When the initial fuel concentration is 750 ppm, as presented in the figures below, the simulated temperature change is greater, between 70 and 90 K, making the utilization of temperature-dependent absorption cross sections even more important for the purpose of obtaining accurate species profiles.

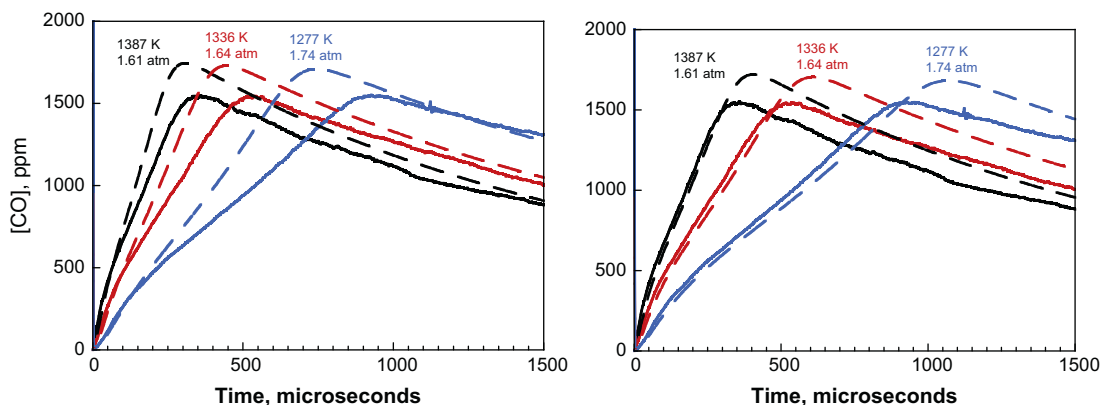


Fig. 9. Measured and simulated CO time histories for 400 ppm 3-pentanone with 5600 ppm O₂ in Ar ($\phi = 0.5$). Left: simulation results before core model updates; right: simulation results with current mechanism.

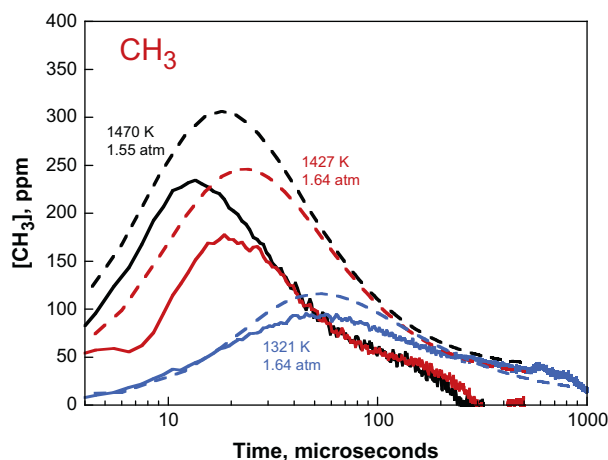


Fig. 10. Measured and simulated methyl time histories for 750 ppm 3-pentanone with 5250 ppm O₂ in Ar ($\phi = 1.5$), plotted on a log-log scale for increased clarity. Simulations with the model prior to and after core chemistry updates are nearly identical.

Fig. 9 below compares experimental and modeled CO profiles under various conditions. The initial CO formation rate is captured reasonably well, although predicted CO peaks are consistently too high by $\sim 10\%$ (equivalent to the experimental uncertainty). CO is produced from just a few reactions for 3-pentanone oxidation under the conditions of this study. At very early times, CO production

is exclusively dominated by the 3-pentanone decomposition followed by the immediate decomposition of propionyl to C₂H₅ + CO (just as it is during pyrolysis). As seen in Fig. 9, this behavior is captured reasonably well, but the CO concentration is underpredicted likely for the same reason as it is during pyrolysis – as stated above, this may be due to too low a branching ratio of k_1/k_2 . Formyl decomposition then dominates CO formation until just after its peak. Finally, the consumption of CO by the reaction CO + OH monotonically increases with time up to the CO peak, and slowly decreases from then on. Although CO production and consumption are governed by these three aforementioned reactions, a sensitivity analysis of CO mole fraction to the rate parameters illustrates that it is sensitive to many core chemistry reactions, thus further refinement of the core H₂/CO/C₁–C₄ chemistry is expected to result in better agreement between the observed and simulated peaks. Because the discrepancy is roughly equivalent to the uncertainty in experimental data, we conclude that the agreement is satisfactory.

The methyl profile comparisons depicted in Fig. 10 exhibit qualitative similarities to those in Fig. 5. The methyl peaks are also overpredicted and the decay is not captured very well. The agreement is less than satisfactory, as the uncertainty all methyl profiles is $\pm 10\%$ during the first 100 μ s, and $\pm 20\%$ at later times. That the model overpredicts the data also suggests a higher branching ratio of k_1/k_2 , and this hypothesis will be explored in future work through the use of reliable quantum chemistry and rate theory calculations. On the other hand, the H₂O profiles presented in Fig. 11 show good agreement with the simulation results. The only nota-

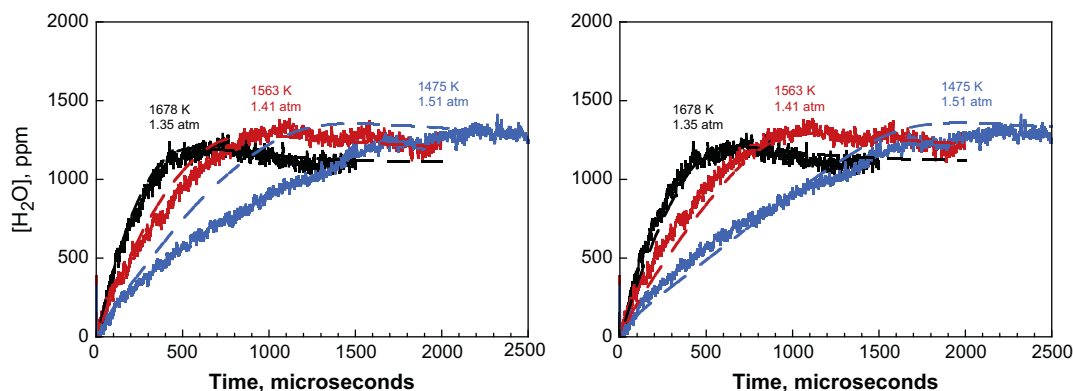


Fig. 11. Measured and simulated H₂O time histories for 400 ppm 3-pentanone with 1870 ppm O₂ in Ar ($\phi = 1.5$). Left: simulation results before core model updates; right: simulation results with current mechanism.

ble discrepancy is at 1475 K, where the observed H_2O profile approaches plateau level with a softer degree of curvature than the simulation results. Nevertheless, the model does agree with the data to within its experimental uncertainty of 10%.

Fig. 12 presents OH and H_2O profiles for 750 ppm of 3-pentanone in argon with unity equivalence ratio. As was the case for the aforementioned shock tube oxidation data, the core chemistry updates resulted in significantly improved agreement between the model predictions and experimental data. Early OH rise is captured significantly better than before the core chemistry changes, as illustrated by the insets of Fig. 12. The observation that the simulated first OH peak is higher than the measured peak for the lower temperatures and lower than the measured peak at higher temperature may suggest the activation energies of R1 and R2 are incorrect. This would not come as a surprise since the rates were estimated from the reverse direction using analogous reactions and computed using thermodynamics. In general, the model-data agreement is worse for the 750 ppm and unity equivalence ratio data compared with the 400 ppm oxidation data. H_2O rise is predicted to be too slow for all cases. Agreement between the model and data are generally better for the OH profiles. The worst agreement is highlighted by the 1349 K OH profile, where the time of the second OH rise is over-predicted by about 200 μs , or 20%. This disagreement is not necessarily due to deficiencies in the model. Recall that the measured temperature uncertainty is roughly 1%. Increasing the initial temperature in the aforementioned simulation by 13.5 K results in excellent agreement of the model prediction with the both the OH and H_2O species profiles, as shown in Fig. 13. That the model-data agreement is excellent for many other cases suggests that the reported postshock temperature for this particular case may be closer to that of its higher uncertainty limit.

Lastly, Fig. 14 compares measured [7] ignition delay times with those simulated using the current mechanism. The shock conditions span a wide range of equivalence ratios, from 0.50 to 2.00. The agreement is satisfactory – within a factor of two.

Kinetic rate parameter uncertainties can be propagated into the shock tube simulation results through a Monte Carlo sampling of the kinetic rate parameter uncertainty space. This approach has been taken in a recent study of n-heptane combustion kinetics which also includes model comparisons against species profiles behind reflected shock waves [56]. Because the base mechanism used in this work and that of Sheen and Wang are nearly identical, and because the dominant reactions contributing to OH and H_2O species profiles are also very similar, a Monte Carlo sampling of the kinetic rate parameter uncertainty space for the conditions of this work will result in similar results to those shown by Sheen and Wang [56].

It is important to reiterate the influence and importance of the core chemistry on the oxidation kinetics of 3-pentanone, its intermediates, and stable product species that are created during the oxidation process. Recall that good agreement between simulations and the pyrolysis data were achieved by addressing the 3-pentanone pyrolysis kinetics [as well as reducing the ethyl decomposition rate by 40%]. However, agreement between the model and oxidation data presented here was only improved through addressing the smaller hydrocarbon chemistry. Clearly, the core chemistry of JetSurF 2.0 is in need of such updates. Although significant improvements were achieved through incorporating the changes listed in Table 3, the discussion above illustrates there is still room for further improvement, which is beyond the scope of the current work.

The effort necessary for the development of an accurate detailed hydrocarbon chemical model is best exemplified through GRI-

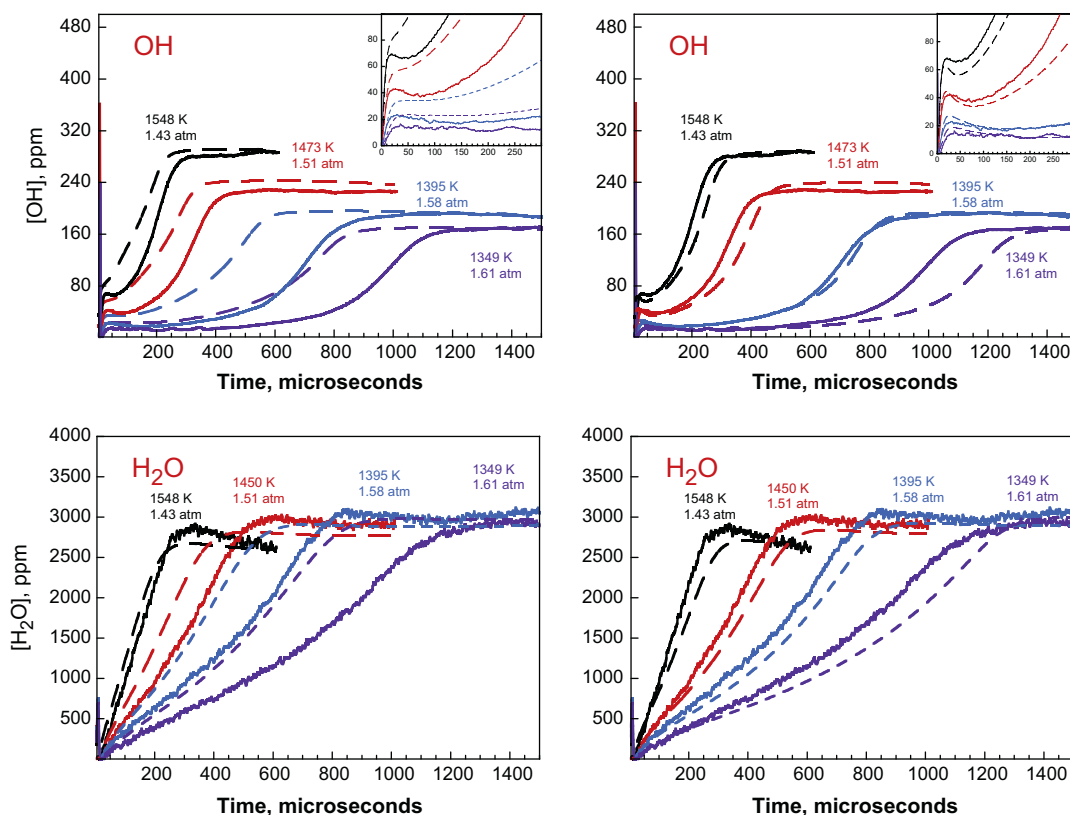


Fig. 12. Measured and simulated OH and H_2O time histories for 750 ppm 3-pentanone with 5250 ppm O_2 in Ar ($\phi = 1.0$). Left: simulation results before core model updates; right: simulation results with current mechanism.

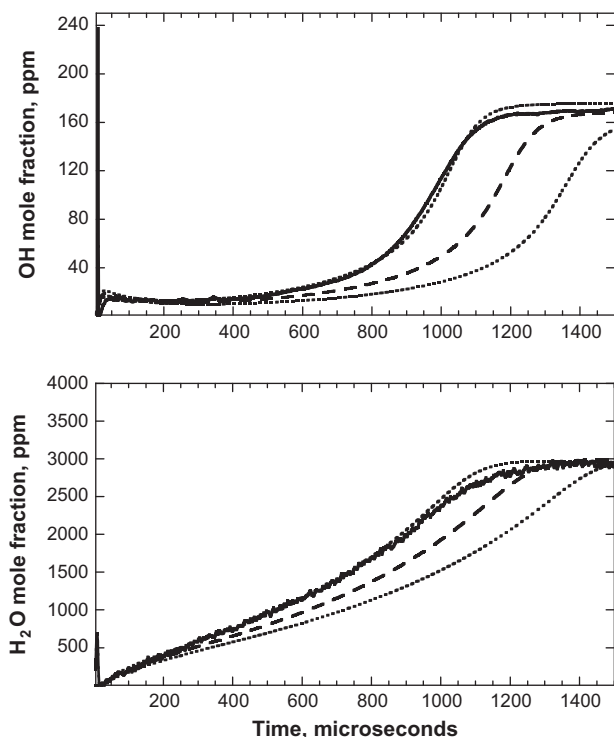


Fig. 13. Measured and simulated OH and H₂O time histories for 750 ppm 3-pentanone with 5250 ppm O₂ in Ar ($\phi = 1.0$). Solid lines: experimental data; long dashed lines: nominal simulation results with current mechanism; dotted lines: simulation results with initial temperature changed by $\pm 1\%$.

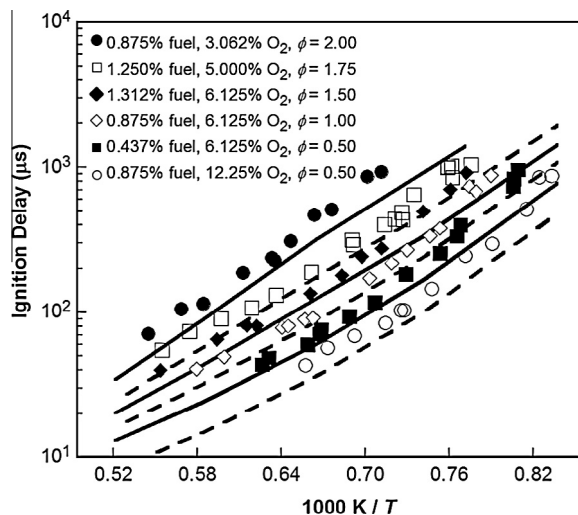


Fig. 14. Simulated (lines) and measured (symbols) [7] ignition delay times for 3-Pentanone (fuel) – O₂ – Argon mixtures with post-shock pressures of 1 atm. Mixture compositions correspond to symbols in the same sequential order depicted.

Mech 3.0, describing the oxidation of methane and natural gas [41]. This model required detailed chemistry for H₂, CO, and methane (it also included NO and limited C₂ chemistry). Molecular growth kinetics leading to aromatic molecules was neglected. Nevertheless, the development effort involved over 15 contributors and took a decade. The model, composed of 325 elementary reactions, was tested against over 100 experimental targets. Most of the original rate constants used were either determined experimentally or by combined quantum chemistry – reaction rate

theory calculations. The model was optimized within the attendant uncertainty of each rate parameter for roughly 1000–2500 K, 10 torr to 10 atm, and equivalence ratio from 0.1 to 5 [41]. Assembly of GRI-Mech was not a trivial task. Even with such large-scale effort, it was impossible to ensure that the submodel (e.g., H₂/CO) was predictive for the combustion of syngas, as evidenced by recent work [57,58]. Because this small hydrocarbon chemistry governs the overall kinetics of the oxidation of larger hydrocarbons under many conditions, it remains crucial to incorporate accurate core chemistry into models for larger systems.

4. Conclusions

Laser-based OH, CO, CH₃, and H₂O species histories during 3-pentanone oxidation were measured behind reflected shock waves over the temperature range of 1277–1678 K at pressures around 1.6 atm, and for equivalence ratios of 0.5, 1.0 and 1.5. The C³ mechanism [7] and insight from our previous work have been utilized to develop an improved kinetic mechanism for the pyrolysis and oxidation of 3-pentanone, consisting of 13 species and 61 reactions. The 3-pentanone unimolecular decomposition rate was increased from its nominal value, and the ethyl decomposition rate decreased from its nominal value [30], but still within its uncertainty of $\pm 50\%$. These two changes resulted in good agreement with our previously reported pyrolysis data, although methyl mole fraction peaks are still overpredicted. Reliable thermodynamic parameters for the radical species within the 3-pentanone submechanism were also determined. Lastly, several updates to the H₂/CO/C₁–C₄ base mechanism of JetSurF 2.0 [23] were made and found to result in significant improvements of model predictions with shock tube oxidation data.

The decomposition of C₂H₅COC₂H₄-s (R3) in the work of Serinyel et al. [7] was discovered to be too high by several orders of magnitude. This resulted in a simulated excess of methyl ketene. An error in the estimation for the rate of methyl ketene decomposition in our previous work led to the incorrect conclusion that methyl ketene unimolecular decomposition is necessary in 3-pentanone kinetics. However, with a proper estimation of methyl ketene decomposition, its decay through unimolecular decomposition is found to be negligible and primarily occurs through bimolecular reaction with gas phase radical species, in contrast to our previous conclusions [8]. Lastly, missing yet important pathways were introduced into the 3-pentanone submechanism: the H-shift reaction of C₂H₅COC₂H₄-p $\leftrightarrow$ C₂H₅COC₂H₄-s and H-elimination from C₂H₅COC₂H₄-s.

The overall agreement between the model and the data presented here is satisfactory, and in many cases, the agreement is excellent. Agreement between the model and higher fuel concentration (750 ppm) oxidation data could be improved, possibly through further updates in the core chemistry. However, minor improvements between the present model and shock tube data are expected through a more reliable determination of 3-pentanone competitive branching kinetics, something that is planned in the near future. Also planned is the utilization of this mechanism to test an experimental design methodology [10] and determine the minimal basis set of experimental data needed to constrain the model rate parameter uncertainties to the same degree as the full experimental data set.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combust-flame.2013.11.010>.

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